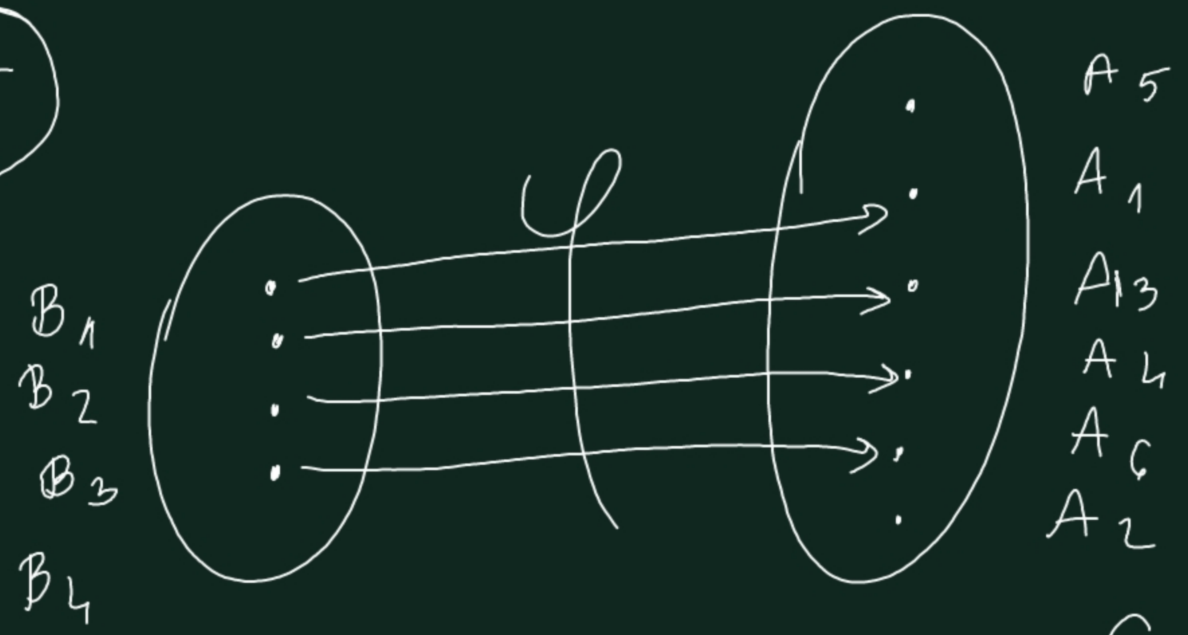
Osmetlés

Altér



$$\varphi = \{ (1,1), (2,3), (3,4), (4,6) \}$$

$m = 4$

$[1..m] \to [1..n]$

Egy adott poz. term. számhoz adjuk meg a hozzá legközelebb eső prímszámot!

(Legyen a „prím" függvény alapértelmezetten ismert.)

$$A = (n : \mathbb{N}^+,\ prím : \mathbb{N}^+)$$

$$F = \left\{ (a,b) \in A \times A \;\middle/\; \begin{array}{l} n(a) = n(b) \wedge prím(prím(b)) \wedge \\ \forall szám \in \mathbb{N}^+ : \\ |n(a) - szám| < \\ |n(a) - prím(b)| \longrightarrow \neg prím(szám) \end{array} \right\}$$

1. feladatsor / 3. / c

$F \subseteq A \times A$

$A = (m : \mathbb{N}, n : \mathbb{N}, p : \mathbb{N})$

$\{m : 20, n : 28, p : 12\} \longrightarrow \{m : 20, n : 28, p : 29\}$

20 28

23 5

29 30 31 32

$\{m : 20, n : 28, p : 12\} \rightarrow \{m : 20, n : 28, p : 31\}$

$\{m : 20, n : 24, p : 12\}$ ... semmit $F(\{m : 20, n : 24, p : 12\}) = \emptyset$

20 23 24 21

$\{m:33,\ n:36,\ p:12\} \rightarrow$ ...

végtelen

0 prímszám

Legyen A egy tetszőleges ábécé,

$S \subseteq A \times (\bar{A} \cup \{fail\})^{**}$ program.

S program programfüggvénye $p(S) \subseteq A \times A$, amennyiben

(1) $D_{p(S)} = \{a \in A \,/\, S(a) \subseteq \bar{A}^*\}$

(2) $\forall a \in D_{p(S)}: p(S)(a) = \{b \in A \,/\, \exists \alpha \in S(a): \alpha_{|\alpha|} = b\}$

$$\underline{\underline{D_{P(S)}}} = \{ a \in A \,/\, S(a) \subseteq \bar{A}^* \} = \underline{\underline{\{2,4\}}}$$

$$p(S)(4) = \{2,1\}$$

$$p(S)(3) = \emptyset$$

$$\underline{\underline{p(S) = \{(2,1),(2,4),(4,2),(4,1)\}}}$$

gyenge programfüggvény

Legyen A ~~tetszőleges~~ állapotok.

Legyen $S \subseteq A \times (\bar{A} \cup \{fail\})^{**}$ program.

S gyenge programfüggvénye $\tilde{p}(S) \subseteq A \times (A \cup \{fail\})$,

amennyiben

1. $D_{\tilde{p}(S)} = \{a \in A \mid S(a) \cap (\bar{A} \cup \{fail\})^* \neq \emptyset\}$

2. $\forall a \in D_{\tilde{p}(S)} : \tilde{p}(S)(a) = \{b \in A \cup \{fail\} \mid \exists \alpha \in S(a) \cap (\bar{A} \cup \{fail\})^* : \alpha_{|\alpha|} = b\}$

$$\tilde{\varphi}(S) = \{(1,1), (1,2), (2,1), (2,4), (4,2), (4,1), (5,4), (5, \text{fail})\}$$

$-5 \ldots$ fail	$1 \ldots 10$	$11 \ldots$	$\varphi(S) =$
$-4 \ldots$ Fail	$2 \ldots 10$	$12 \ldots \infty$	$= \{(1, 10),$
$-3 \ldots$	3	$\vdots$	$(2, 10),$
$-2 \ldots$ fail	4		$(3, 10), \ldots,$
$-1 \ldots$	5		$(10, 10)\}$
$0 \ldots 0, 0, 0, \ldots$	6		
	7	10	
	8		
	$9 \ldots$		
	10		

$$\tilde{\varphi}(S) = \varphi(S) \cup \{(a, \text{fail}) \mid -5 \leq x(a) \leq -1, \ a \in A\}$$

Megoldás

Legyen A tetszőleges állapottér.

Legyen $S \subseteq A \times (\bar{A} \cup \{\text{Fail}\})^{**}$ program,

$F \subseteq A \times A$! S megoldja az F-et, ha

1. $D_F \subseteq D_{\varphi(S)}$
2. $\forall a \in D_F : \varphi(S)(a) \subseteq F(a)$

Rajz

	a
1	☒
2	☒
3	☐
4	☐

	b
1	☒
2	☐

	c
1	☐
2	☒
3	☐
4	☒

	d
1	☒

$$F = \{(a,1),(a,2),(a,3),(a,4),(b,1),(b,2),(c,1),\dots,(c,4),(d,1)\}$$

$D_F \subseteq D_{p(S)}$

$\{2,4\} \subseteq \{2,4\}$ ✓

Igen.

$\forall a \in D_F: p(S)(a) \subseteq F(a)$

$p(S)(2) = \{1,4\} \subseteq F(2) = \{1,4\}$ ✓

$p(S)(4) = \{2,1\} \subseteq F(4) = \{1,2,5\}$ ✓

3. $A = \{1, 2, 3\}$

$F = \{(1,1), (1,2), (2,3)\}$

$S = \{ 1 \to \langle 1, 2 \rangle$

$2 \to \langle 2, 3 \rangle$

$3 \to \langle 3, 3 \rangle$

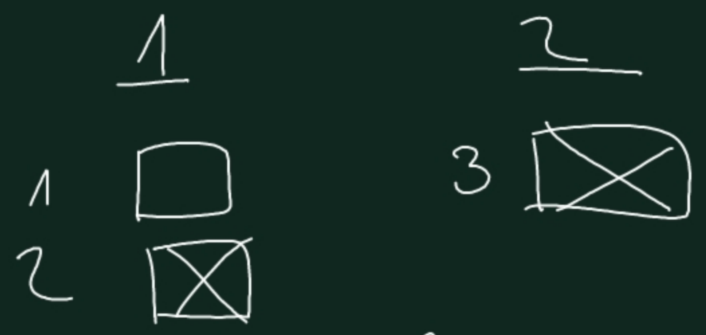
$3 \to \langle 3, \text{fail} \rangle$

$3 \to \langle 3, 3, \dots \rangle$

$\}$

$\to$ ne je kategorizirana

Rajz



$\varphi(S) = \{(1,2), (2,3)\}$

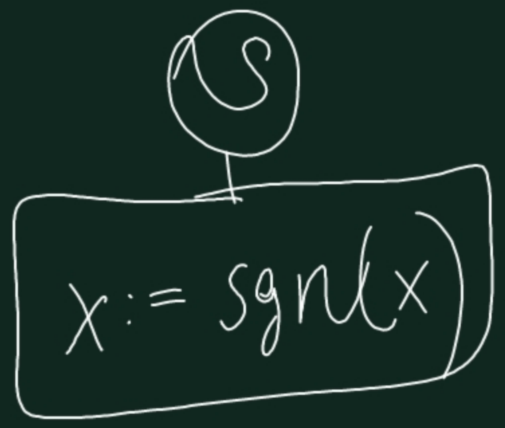
S_2 A fellettri program.

$S_2 = S$

Pluszpont

$$A = (x : \{2, 3, 4, 5\})$$

S program az A felett.



Adjuk meg S-et!